
TP : 01

Exercise 1 :

Objective:

Understand the limitations of representing real numbers on a computer and their consequences for numerical calculations.

Algorithm 1 Machine Precision Algorithm

```
1: Initialize  $\epsilon \leftarrow 1.0$ 
2: while  $(1 + \epsilon > 1)$  do
3:    $\epsilon \leftarrow \epsilon / 2.0$ 
4: end while
5:  $\epsilon \leftarrow \epsilon \times 2.0$ 
6: Machine precision:  $\epsilon_m = \epsilon$ 
```

1. Write a MATLAB script that determines the machine precision ϵ_m , equivalent to the following algorithm.
2. Comparer la valeur obtenue avec la fonction MATLAB intégrée : $\epsilon_{matlab} = \epsilon$;
3. What is the difference? Why ?.
4. give results in single precision and double precision (Standard IEEE-754).

Exercise 2 :

Objective:

To understand how rounding errors can affect numerical computations and how the **order of operations** can change the final result due to **floating-point arithmetic limitations**.

Given:

$$(x + y) + z \quad \text{and} \quad x + (y + z)$$

with:

```
x = 0.854e3;
y = 0.251e3;
z = 0.852e3;
```